

Lecture 36

STAT 109: Introductory Biostatistics - Simple linear regression (part 2)

Lecture 36

Inference in simple linear regression

If the linear model is appropriate and residuals are roughly normal with constant spread, we can do interval estimation:

- Confidence interval for the slope `beta1`.
- Confidence interval for the mean response $E(Y \mid X = x_0)$.
- Prediction interval for an individual future Y at x_0 .

The prediction interval is wider than the mean-response confidence interval because it includes both uncertainty in the mean and individual-to-individual variation.

Learning objectives

By the end of this lecture, you should be able to:

- explain when normal-residual assumptions support interval inference in linear regression;
- compute and interpret a confidence interval for slope `beta1`;
- compute and interpret a confidence interval for the mean response at x_0 ;
- compute and interpret a prediction interval for an individual response at x_0 ;
- explain why prediction intervals are wider than confidence intervals for means;
- report test set MSE and test R^2 as primary predictive performance summaries.

Example 1 - Study hours and exam score (continued)

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {  
  install.packages("tidyverse", repos = "https://cloud.r-project.org")  
}  
  
library(tidyverse)  
  
set.seed(3501)  
n <- 160  
exam_dat <- tibble(  
  study_hours = runif(n, 1, 15),  
  exam_score = 45 + 3.4 * study_hours + rnorm(n, 0, 8)  
)
```

Train/test split and fit

```
idx_train <- sample(seq_len(nrow(exam_dat)), size = 0.8 * nrow(exam_dat))  
exam_train <- exam_dat[idx_train, ]  
exam_test  <- exam_dat[-idx_train, ]
```

```
fit_exam <- lm(exam_score ~ study_hours, data = exam_train)
summary(fit_exam)
```

Call:

```
lm(formula = exam_score ~ study_hours, data = exam_train)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-16.0920	-4.3112	-0.8295	4.4390	21.3982

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	42.4805	1.3406	31.69	<2e-16 ***
study_hours	3.8436	0.1551	24.78	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.664 on 126 degrees of freedom

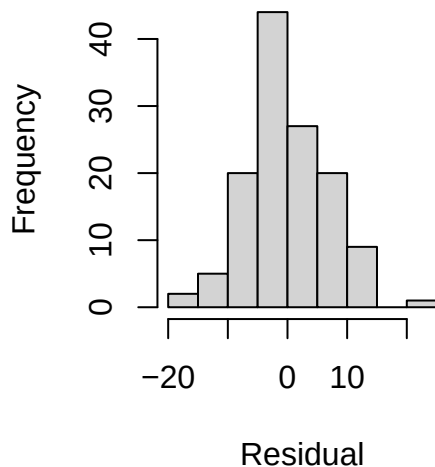
Multiple R-squared: 0.8298, Adjusted R-squared: 0.8284

F-statistic: 614.2 on 1 and 126 DF, p-value: < 2.2e-16

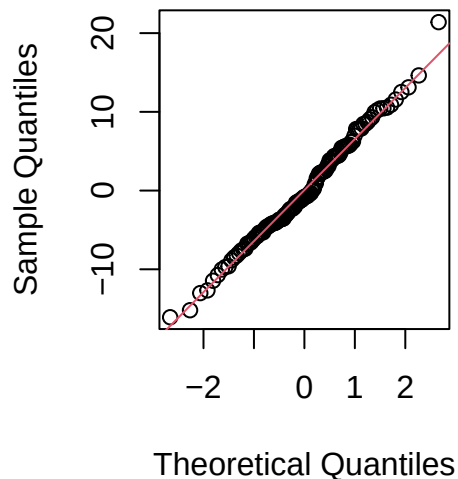
Residual normality check (visual)

```
par(mfrow = c(1, 2))
hist(residuals(fit_exam), main = "Residual histogram", xlab = "Residual")
qqnorm(residuals(fit_exam)); qqline(residuals(fit_exam), col = 2)
```

Residual histogram



Normal Q-Q Plot



```
par(mfrow = c(1, 1))
```

1) CI for slope β_1

```
confint(fit_exam, "study_hours", level = 0.95)
```

```
          2.5 %    97.5 %  
study_hours 3.536713 4.150568
```

Interpretation pattern: we are 95% confident that each additional study hour changes mean exam score by a value in this interval.

2) CI for mean response at x_0

For $x_0 = 10$ study hours:

```
x0 <- tibble(study_hours = 10)  
predict(fit_exam, newdata = x0, interval = "confidence", level = 0.95)
```

```
      fit      lwr      upr  
1 80.91687 79.56421 82.26953
```

Interpretation: 95% CI for the mean exam score among all students who study 10 hours.

3) Prediction interval for an individual response at x_0

```
predict(fit_exam, newdata = x0, interval = "prediction", level = 0.95)
```

```
      fit      lwr      upr  
1 80.91687 67.65902 94.17473
```

Interpretation: 95% interval for a single future student's score if they study 10 hours.

Training vs test performance reminder

```
mse <- function(y, yhat) mean((y - yhat)^2)  
  
train_pred <- predict(fit_exam, newdata = exam_train)  
test_pred  <- predict(fit_exam, newdata = exam_test)  
  
train_mse <- mse(exam_train$exam_score, train_pred)  
test_mse  <- mse(exam_test$exam_score, test_pred)  
  
train_r2 <- 1 - sum((exam_train$exam_score - train_pred)^2) /  
  sum((exam_train$exam_score - mean(exam_train$exam_score))^2)  
test_r2  <- 1 - sum((exam_test$exam_score - test_pred)^2) /  
  sum((exam_test$exam_score - mean(exam_test$exam_score))^2)  
  
tibble(  
  set = c("train", "test"),  
  MSE = c(train_mse, test_mse),  
  R2 = c(train_r2, test_r2)  
)
```

```
# A tibble: 2 x 3  
  set      MSE      R2  
<chr> <dbl> <dbl>
```

```
1 train  43.7 0.830
2 test   66.3 0.767
```

Use test metrics as your main generalization report:

- Test MSE = typical squared prediction error on new cases.
 - Test R^2 = proportion of test-set variation explained by the linear model.
-

Example 2 - Fertilizer dose and plant height

```
set.seed(3502)
plant_dat <- tibble(
  dose = runif(95, 0, 12),
  height_cm = 18 + 2.2 * dose + rnorm(95, 0, 5.3)
)

idx <- sample(seq_len(nrow(plant_dat)), size = 0.8 * nrow(plant_dat))
plant_train <- plant_dat[idx, ]

fit_plant <- lm(height_cm ~ dose, data = plant_train)
summary(fit_plant)
```

Call:

```
lm(formula = height_cm ~ dose, data = plant_train)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-10.7136	-3.3558	0.5898	3.7855	8.9421

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	18.3691	1.3412	13.70	<2e-16 ***
dose	2.1394	0.1706	12.54	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.961 on 74 degrees of freedom

Multiple R-squared: 0.68, Adjusted R-squared: 0.6756

F-statistic: 157.2 on 1 and 74 DF, p-value: < 2.2e-16

```
# CI for slope beta1
confint(fit_plant, "dose")
```

	2.5 %	97.5 %
dose	1.799457	2.479396

```
# CI for mean response at x0 = 7
new_x <- tibble(dose = 7)
predict(fit_plant, newdata = new_x, interval = "confidence")
```

	fit	lwr	upr
--	-----	-----	-----

```
1 33.34505 32.2105 34.4796
```

```
# Prediction interval for an individual at x0 = 7
predict(fit_plant, newdata = new_x, interval = "prediction")
```

```
      fit      lwr      upr
1 33.34505 23.3956 43.2945
```

Example 3 - Exercise minutes and resting heart rate

```
set.seed(3503)
heart_dat <- tibble(
  exercise_min = runif(110, 0, 60),
  resting_hr = 79 - 0.30 * exercise_min + rnorm(110, 0, 4.8)
)

idx <- sample(seq_len(nrow(heart_dat)), size = 0.8 * nrow(heart_dat))
heart_train <- heart_dat[idx, ]

fit_heart <- lm(resting_hr ~ exercise_min, data = heart_train)
summary(fit_heart)
```

Call:

```
lm(formula = resting_hr ~ exercise_min, data = heart_train)
```

Residuals:

Min	1Q	Median	3Q	Max
-11.6123	-3.1631	0.3452	2.9383	14.2460

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	80.00362	1.17719	67.962	< 2e-16 ***
exercise_min	-0.31807	0.03237	-9.827	1.02e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.854 on 86 degrees of freedom

Multiple R-squared: 0.5289, Adjusted R-squared: 0.5235

F-statistic: 96.57 on 1 and 86 DF, p-value: 1.023e-15

```
# CI for slope beta1
confint(fit_heart, "exercise_min")
```

	2.5 %	97.5 %
exercise_min	-0.3824177	-0.2537277

```
# CI for mean resting HR at x0 = 30 min
new_x <- tibble(exercise_min = 30)
predict(fit_heart, newdata = new_x, interval = "confidence")
```

```
      fit      lwr      upr
1 70.46144 69.41853 71.50435
```

```
# Prediction interval for one person at x0 = 30 min
predict(fit_heart, newdata = new_x, interval = "prediction")
```

```
      fit      lwr      upr
1 70.46144 60.75523 80.16766
```

Key takeaways

- Slope CI answers: what plausible values for the linear effect **beta1** are supported by data?
- Mean-response CI answers: where is the average outcome for all units at **x0**?
- Prediction interval answers: where might one future individual outcome fall at **x0**?
- Keep reporting test MSE and test R^2 for predictive performance on new data.